

Education Knowledge Base - Part 2: AI & ML Track

Format: PDF | Covers: Data Structures, Algorithms, Machine Learning, Deep Learning, Artificial Intelligence

Course: Data Structures

Overview

Data Structures are ways of organizing and storing data so it can be accessed and modified efficiently.

Mastering data structures is fundamental to writing efficient programs and is one of the most tested topics in technical interviews.

Duration

The Data Structures course typically takes 8 weeks. Consistent practice on platforms like LeetCode

accelerates learning.

Prerequisites

Before starting Data Structures, a learner should have: knowledge of at least one programming language

such as Python, Java, or C++.

Key Topics Covered

Arrays, linked lists, stacks, queues, hash tables, trees (binary, BST, AVL), heaps, graphs, and tries.

Learning Resources

Recommended resources: Introduction to Algorithms (CLRS), GeeksforGeeks Data Structures section,

and the 'Grokking Algorithms' book.

Recommended Next Course

After Data Structures, the next step is Algorithms. Algorithms build directly on data structures to solve

complex problems efficiently.

Career Opportunities

Roles include Software Engineer, Backend Developer, and Competitive Programmer. Strong data

structure knowledge is required to clear interviews at top tech companies.

Real World Applications

Data structures underpin databases, search engines, routing algorithms, and virtually all software systems.

Course: Algorithms

Overview

Algorithms are step-by-step procedures for solving computational problems. Studying algorithms teaches

you how to design efficient solutions and analyze their time and space complexity, which is central to

computer science and software engineering.

Duration

The Algorithms course typically takes 8 weeks. Understanding complexity analysis (Big-O) is the key milestone.

Prerequisites

Before starting Algorithms, a learner should have: solid grasp of Data Structures.

Key Topics Covered

Sorting and searching, recursion, divide and conquer, greedy algorithms, dynamic programming, graph algorithms (BFS, DFS, Dijkstra), and complexity analysis.

Learning Resources

Recommended resources: Introduction to Algorithms (CLRS), Algorithms by Robert Sedgewick, and the

MIT OpenCourseWare Algorithms course.

Recommended Next Course

After Algorithms, the next step is Machine Learning. Algorithmic thinking is the foundation for understanding machine learning models.

Career Opportunities

Roles include Software Engineer, Research Engineer, and Machine Learning Engineer.

Algorithm mastery

separates strong engineers from average ones in interviews and real work.

Real World Applications

Algorithms drive GPS navigation, recommendation systems, compression, cryptography, and search ranking.

Course: Machine Learning

Overview

Machine Learning (ML) is a branch of artificial intelligence where systems learn patterns from data to make predictions or decisions without being explicitly programmed. It powers recommendation engines, fraud detection, forecasting, and much of modern AI.

Duration

The Machine Learning course typically takes 10 weeks. A strong math foundation significantly speeds up understanding.

Prerequisites

Before starting Machine Learning, a learner should have: Python and Mathematics (linear algebra, probability, and statistics).

Key Topics Covered

Supervised learning (regression, classification), unsupervised learning (clustering, dimensionality reduction), model evaluation, feature engineering, overfitting and regularization, and popular libraries like

scikit-learn.

Learning Resources

Recommended resources: Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow by

Aurelien Geron, Andrew Ng's Machine Learning course, and the scikit-learn documentation.

Recommended Next Course

After Machine Learning, the next step is Deep Learning. Deep Learning extends machine learning to

neural networks for complex tasks like vision and language.

Career Opportunities

Roles include Machine Learning Engineer, Data Scientist, and AI Researcher. ML skills are among the

highest paid and most in-demand in the tech industry.

Real World Applications

ML powers Netflix recommendations, spam filters, credit scoring, demand forecasting, and predictive maintenance.

Course: Deep Learning

Overview

Deep Learning is a subfield of machine learning based on artificial neural networks with many layers. It

excels at learning from large, unstructured data such as images, audio, and text, and is the technology

behind modern computer vision and large language models.

Duration

The Deep Learning course typically takes 12 weeks. Access to GPUs and practical projects greatly

improves outcomes.

Prerequisites

Before starting Deep Learning, a learner should have: Machine Learning fundamentals and Python.

Key Topics Covered

Neural networks, backpropagation, convolutional neural networks (CNNs), recurrent neural networks

(RNNs), transformers, attention mechanisms, and frameworks like PyTorch and TensorFlow.

Learning Resources

Recommended resources: Deep Learning by Ian Goodfellow, the [fast.ai](#) course, Andrew Ng's Deep

Learning Specialization, and the PyTorch documentation.

Recommended Next Course

After Deep Learning, the next step is Artificial Intelligence. A broader AI study ties deep learning together

with reasoning, agents, and generative systems.

Career Opportunities

Roles include Deep Learning Engineer, Computer Vision Engineer, and NLP Engineer. Deep learning expertise is central to cutting-edge AI products like ChatGPT and self-driving cars.

Real World Applications

Deep learning powers facial recognition, medical image diagnosis, speech recognition, and generative AI like GPT and DALL-E.

Course: Artificial Intelligence

Overview

Artificial Intelligence (AI) is the broad field of building systems that can perform tasks requiring human-like intelligence, such as reasoning, perception, planning, and language understanding. It encompasses machine learning, deep learning, generative AI, and agentic systems.

Duration

The Artificial Intelligence course typically takes 12 weeks. AI is a wide field; focusing on a specialization like NLP or generative AI is recommended.

Prerequisites

Before starting Artificial Intelligence, a learner should have: Machine Learning and Python.

Key Topics Covered

Search and planning, knowledge representation, reasoning under uncertainty, natural language processing, generative AI, large language models, retrieval-augmented generation (RAG), and AI agents.

Learning Resources

Recommended resources: Artificial Intelligence: A Modern Approach by Russell and Norvig, OpenAI documentation, LangChain and LangGraph documentation, and the Hugging Face courses.

Recommended Next Course

After Artificial Intelligence, the next step is Specialized AI tracks such as Generative AI, RAG systems, or Agentic AI. From here you can specialize in building RAG pipelines, multi-agent systems, and MLOps for production AI.

Career Opportunities

Roles include AI Engineer, Generative AI Engineer, NLP Engineer, and AI Research Scientist. AI

engineering skills, especially around LLMs and agents, are the fastest growing in the industry.

Real World Applications

AI drives chatbots, autonomous agents, recommendation systems, virtual assistants, and generative content tools.